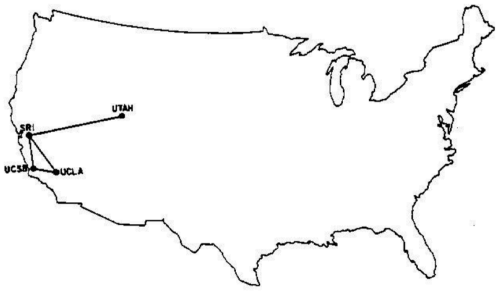
1) List ten important milestones of the internet technology evolution.

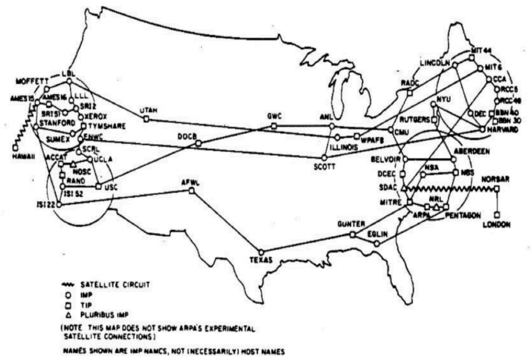
Date	Event
1986	Establishment of the NSFNET (56 kbps)
1987	No. of Internet nodes > 10 k
1988	Establishment of the CERT (Computer Emergency Response Team) by the DARPA
1990	ARPANET = Internet, (No. of nodes > 100 k)
1991	Tim Berners-Lee: invention of the World Wide Web (WWW, CERN)
1993	Development of the first web browser: Mosaic
1994	Development of the Netscape Navigator (browser)
1997	Establishment of the ARIN (American Registry for Internet Numbers), Starting Internet 2
1999	Internet 2: development of the IPv6 Setting of a purpose: integration of the voice, video and data transmission into a common infrastructure

Date	Event
1970	University of Hawaii: development of ALOHANET
1970’s	Penetration of the digital ICs, digital PCs
1972	Ray Tomlinson: development of the E-mail program
1973	Bob Kahn, Vint Cerf: invention of TCP/IP, connection to Europe of the ARPANET
1974	BBN: development of the Telnet. Establishment of the business version of the ARPANET
1980’s	Penetration of the PCs and Unix based minicomputers
1981	Setting the Internet term: network of networks
1982	ISO develops the OSI model and its protocols (these protocols are not used today, just the model)
1983	TCP/IP becomes general mechanism of the Internet. MILNET is split from the ARPANET.
1984	Establishment of the Cisco Systems, Co. (router SW). Development of the DNS, 1000 nodes.

Date	Event
1960's	Mainframe Computing
1962	Paul Baran: description of the packet switching theory
1967	Larry Roberts: paper in the ARPANET subject
1969	Establishment of the ARPANET: UCLA, UCSB, U-Utah, Stanford



1969



1977

Date	Event
Before 1900	Long distance communications (courier, smoke signs, galvanic/optical telegraph)
1890's	Bell: invention of the telephone, rapid penetration rate of the service
1901	Marconi: first Transatlantic wireless transmission
1920's	AM radio
1939	FM radio
1940's	Invention of the microwave
1947	Shockley, Barden, Brittain: invention of the semiconductor transistor
1948	Claude Shannon: „A Mathemtatical Theory of Communication"
1950's	Invention of the IC
1957	Establishment of the ARPA by DoD.

2) Give four classification criteria of the computer network and give examples of each category.

1) Size of the physical area:

- BAN: Body Area Net., BCI – Brain Computer Interface
- PAN: Personal Area Network
- SOHO: Small Office/ Home Office
- LAN: Local Area Network
- MAN: Metropolitan Area Network
- WAN: Wide Area Network
- GAN/Internet: Global Area Network

2) Transmission rate:

- Classical networks: kbps … Mbps
- High speed networks: 100 Mbps … Tbps

3) Ownership:

- Private Network
- Public Network

4) Mobility:

- Fixed Network
- Mobile Network

3) Explain following mechanisms: signal coding, modulation, multiplexing, demultiplexing, transmission rate, modulation rate (text and figure).

Signal, signal coding, modulation, multiplexing:

Signal coding: Process of mapping the digital data to the analog carrier signal (e.g. voltage, voltage modification. We are discussing digital coding only, but exists analog signal coding mechanism, as well.

Modulation/Demodulation: The data transfer channel can be represented as a frequency band (carrier frequency). The digital data is mapped to the analog carrier signal. One of the parameters (e.g. amplitude, phase, etc.) of the carrier signal is modified in function of the data. The inverse process at the receiver is named demodulation. The MODEM device executes modulation task when transmits and demodulation task when receives data. These two processes can be done simultaneously on two different channels by the modem.

Multiplexing/Demultipexing: Two or more separated transmitted flows on the same channel in the „same" time is the multiplexing. Separation process at the receiver of two or more flows from the channel is named demultiplexing.

Modulation rate:

No. of signal level modification on the channel per unit of time (second).
Measurement unit:
signal change / sec = baud
Measures the signal change speed. It has strong correlation with the transmission rate at a given communication mechanism.

Bigger units:

1 kbaud	1.000 baud				10^3 baud
1 Mbaud	1.000 kbaud	1.000.000 baud			10^6 baud
1 Gbaud	1.000 Mbaud	1.000.000 kbaud	1.000.000.000 baud		10^9 baud
1 Tbaud	1.000 Gbaud	1.000.000 Mbaud	1.000.000.000 kbaud	1.000.000.000.000 baud	10^{12} baud

Data transfer rate (network speed, bit rate, bandwidth):

No. of bits transmitted on the channel per unit of time (second).
Unit of measurement:
bit/seconds,
b/s,
bps.
It is used to count the transmission capacity of the channel.

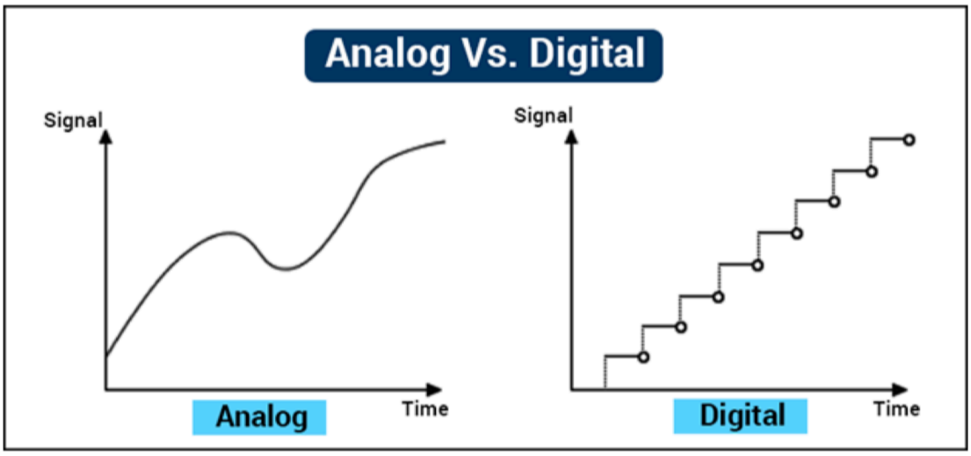
Bigger units:

1 kbps	1.000 bps				10^3 bps
1 Mbps	1.000 kbps	1.000.000 bps			10^6 bps
1 Gbps	1.000 Mbps	1.000.000 kbps	1.000.000.000 bps		10^9 bps
1 Tbps	1.000 Gbps	1.000.000 Mbps	1.000.000.000 kbps	1.000.000.000.000 bps	10^{12} bps

4) Characterize the analog and digital transmission (text and figures).

Signal, signal coding, modulation, multiplexing:

Signal: Energy quantity dependent on space and time. It caries the data on the channel. Variants: analog, digital.



5) List three switching modes and characterize them.

Switching modes:

Circuit switched technology: Dedicated connection path is created between the end nodes before the transfer. This connection is maintained during the session.
E.g.: classic wired phone service

Message switched (store and forward) technology: Circuit is not created between the end nodes, but only on the segment between two consecutive intermediate nodes.
E.g.: Telex

Packet switched technology: The data is fragmented in packets with length < MTU and these units are forwarded separately. The method is efficient and has well manageable buffering features.

6) List four addressing methods and characterize them.

Unicast address: ID of one communication entity (interface). The source ID is unicast address. In general the communication interface has unicast ID, but exceptions exist in function of the communication technology.

Anycast address: ID of a set of communication entities (set of interfaces on different nodes). Message sent to Anycast address should arrive to at least one element of the set (usually the nearest entity).

Multicast address: ID of the set of communication entities (usually set of interfaces on different nodes). Message sent to this ID should arrive to each entity of the set.

Broadcast address: ID of all communication entities in a given (broadcast) domain. I may be interpreted as a special multicast ID, where all entities take part in the communication.

7) Compare three models of the communication and explain the layer functions (text and figures).

OSI Layers:

1. Physical Layer:
Electrical, mechanical, timing and geometrical characteristics of the signals, media and connectors

2. Data link Layer:
Reliable transfer of the L2 PDU (frame) between two nodes connected to the same media (wire, frequency range).
Tasks involved: physical addressing, network topology, media access control, error detection on physical layer, ordered delivery of the frames.
IEEE divided this layer into two sublayers:

- MAC: Medium Access Control sublayer (integrated into the NIC)
- LLC: Logical Link Control sublayer (device driver of the NIC)

3. Network Layer:
Provides connection and path selection (routing) between two nodes.
Tasks involved: logical addressing, routing.

OSI Layers (cont'd):

4. Transport Layer:
Provides reliable connection between two network nodes.
Tasks involved: management of the virtual circuits, detection and correction of the transmission errors, data flow scheduling.

5. Session Layer:
Sets up, schedules and releases the dialogs between the applications (session, dialog control).

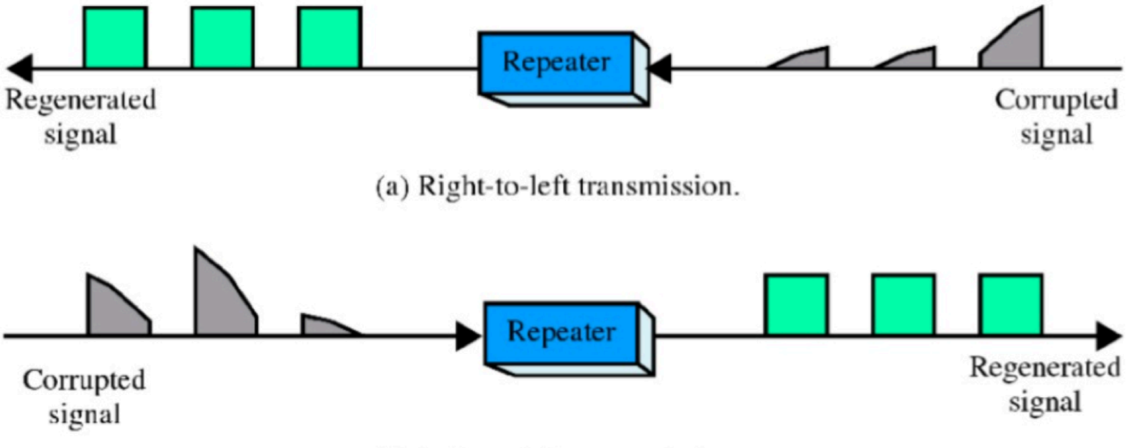
6. Presentation Layer:
Conversion and alignment of the local representation of the data between different computers.

7. Application Layer:
Transfers the content between different application entities (file transfer, e-mail, etc.)

8) List five intermediate nodes and characterize them in detail (text and figure).

Repeater, Hub:

- Active Hub: transmits the signal on the media: reamplification, retiming, resynchronizing, reshaping of the pulses.
- Passive Hub: analogue amplifier without just reamplification.

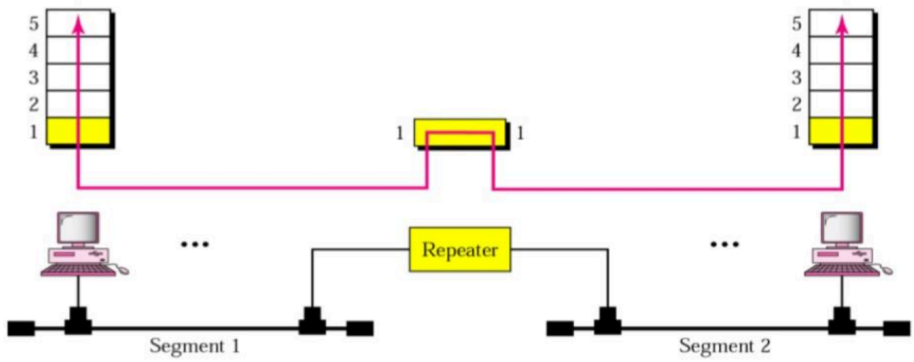


(a) Right-to-left transmission.

(b) Left-to-right transmission.

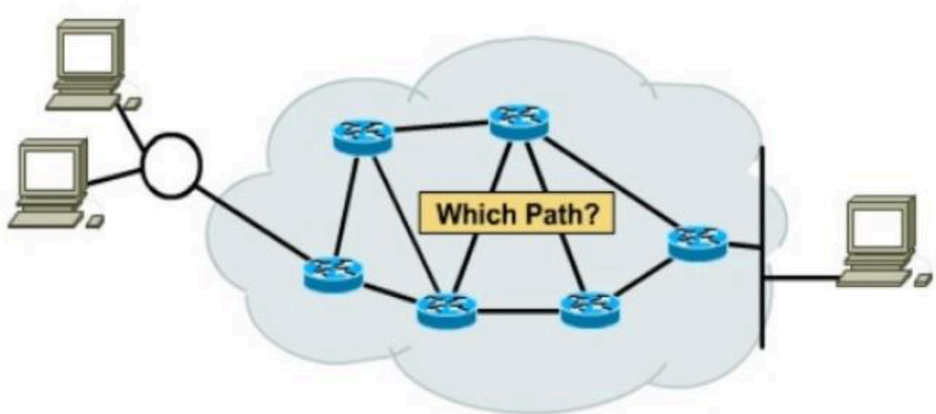
Repeater, Hub (cont'd):

- Connects multiple nodes in L1.
- Transmits the signal on the media: reamplification, retiming, resynchronizing, reshaping of the pulses.

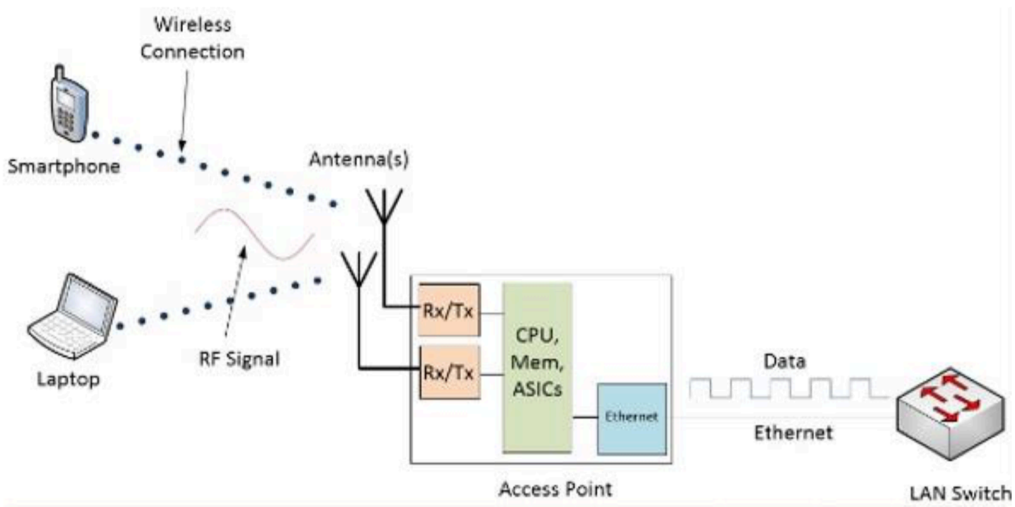


A repeater connects segments of a LAN.
A repeater forwards every frame – there is no filtering.
A repeater is a regenerator, not an amplifier.

Router:
- Connects multiple nodes in L3 and L2 and L1 (broadcast domains).
- Forwards packets, does not forwards frames or bits.
- Executes routing in network layer.
- Each interface has unique physical and unique logical address.

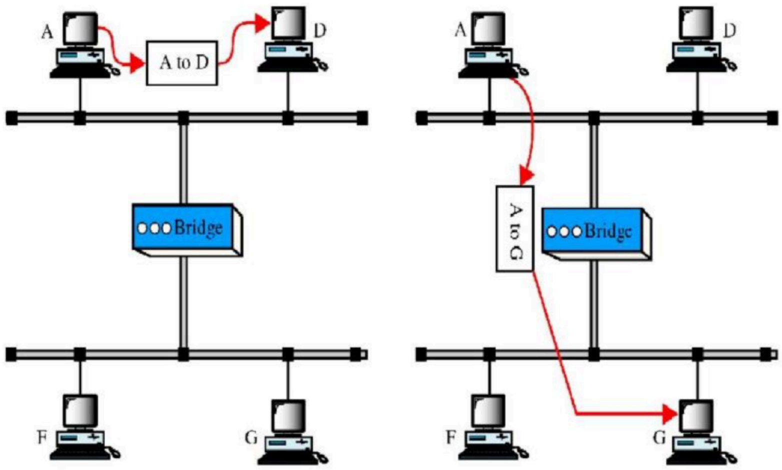


Wireless Access Point (AP):
- Provides special bridging: wired port <-> wireless port.
- Forwards frames, but not bits.
- The frame is transformed during the transmission through the AP



Bridge, Switch:

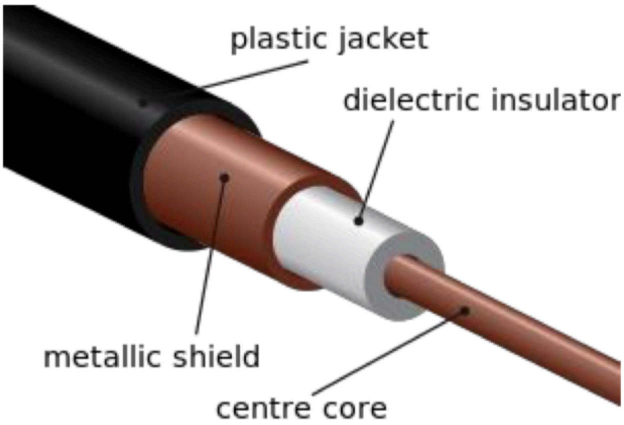
- Connects two or multiple nodes on different media (segments, collision domains) in L1 and L2.
- Learns transparently the source physical address of the devices and transmits the frame only to the corresponding destination node.
- Bridge: service executed by software (one CPU, old device)
- Switch: se



9) Characterize the following network mediums: coaxial cable, fibre optic (text and figures).

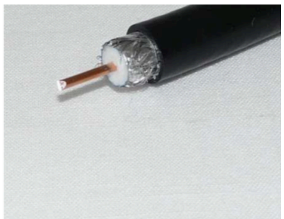
2. Coaxial cable:

- Concentric structure
- Role of the twisting: cancellation of the noises



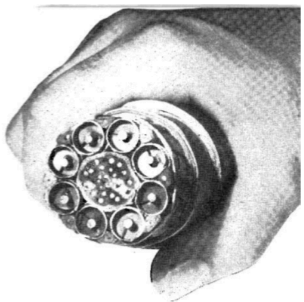
- External diameter size:
  - Thin coaxial cable: 5 mm
  - Thick coaxial cable: 25 mm (thick dielectric insulator)

<u>Wired media (galvanic):</u>

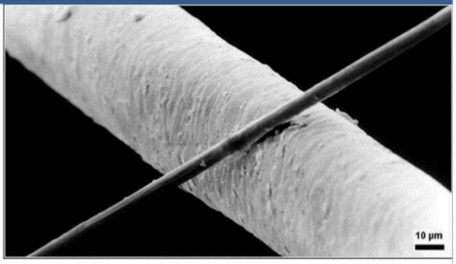


2. Coaxial cable (cont'd):

- Has superior noise immunity (better than the TP)
- Can be used for higher distances, reamplification at n·km
- Has multipoint access usage possibility
- Impedance (frequency dependent):
  - Baseband: 50 Ω (digital signal transmission)
  - Broadband: 75 Ω (analogue signal transmission, CATV)
- Frequency band: ~ 600 MHz
- It was used at the beginning of the LAN services, today TP is used instead:
  - RG-62: 93 Ω (Arcnet, 1970's)
  - RG-58: 50 Ω (Ethernet, 1980's)
- Today is used for antenna connection to the receiver or transmitter.

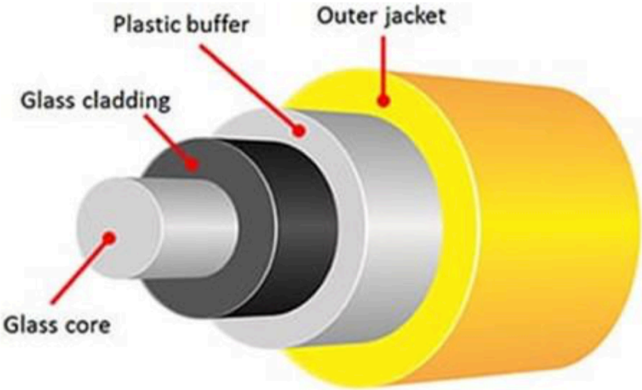


1948 (480 telephone calls, one television channel)

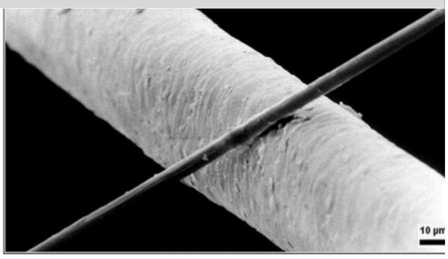


Wired media (optical):

3. Fibre cable:
- Light signal transmission
- Structure:



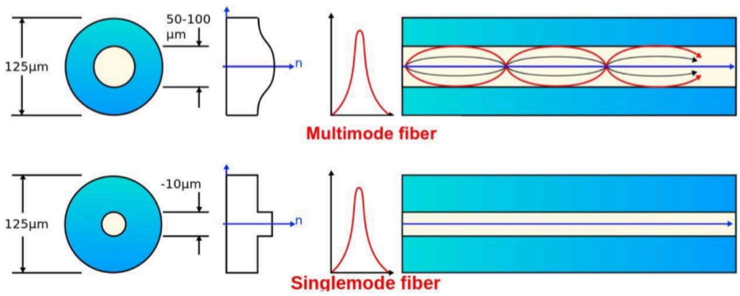
- Core diameter: d
- Cladding diameter: D



Wired media (optical):

3. Fibre cable (cont'd):
- One fibre: multiple optical simultaneous channels
- Usually pair of fibres are used for Tx and Rx
- Light transmission in the fibre
    - MMF (Multimode Fibre)
    - SMF (Single mode Fibre)

	MMF	SMF
D [μm]	125	125
d [μm]	50; 62.5	9.5
λ [μm]	0.85	1.31; 1.55

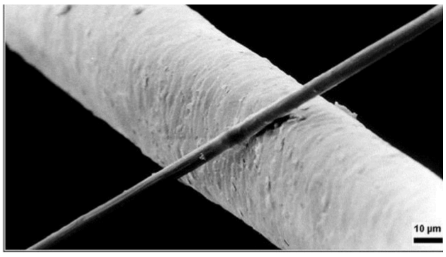


Wired media (optical):

3. Fibre cable (cont'd):

- High transfer rate x distance product:

  $n \cdot 100$ Gbps · km
- Lower mass than the galvanic cables
- Low attenuation, high frequency range
- Exterior EM noises are not collected
- EM noises are not radiated -> signals can not be intercepted
- Low error rate, low maintenance cost.



10) Explain the impact of noise on communication and characterize the SNR term (text and figures).

Noise:

- Sum of the simultaneous energy impulses form other sources than the sender node is the noise.
- If the signal level is comparable with the noise level, then decoding of the information forwarded by the signal is impossible or affected by errors.
- For god reception quality the signal/noise (S/N) ratio should be as high as possible.

Upper limit of the channel transmission rate with noise:

Shannon theorem of the channel upper transmission rate limit

$$C = H \cdot \log_2(1 + S/N)$$

where:

C – upper limit of the transmission rate [bit/sec],
H – bandwidth (frequency domain) of the channel [Hz],
S – Power of the signal [W]
N – Power of the noise [W]

11) Compare unipolar and bipolar coding mechanisms and give two examples of each group (text and figure).

Unipolar coding:

- Two signal level exist: (+V) and (0) symbols.

Bipolar coding:

- Two signal level exist: (+V) and (-V) symbols.

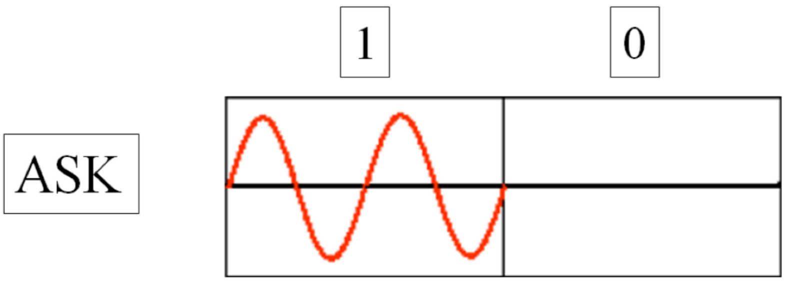
12) List four modulation technics and explain their mechanism (text and figure).

1. Amplitude Shift Keying (ASK) modulation:

„0” bit: absence of the carrier signal ($A = 0$).
„1” bit: presence of the carrier signal ($A \neq 0$).

$$S_{(0)}(t) = 0$$
$$S_{(1)}(t) = A \cdot \sin(\omega_c \cdot t)$$

Simple to execute. Disadvantage is the presence of the constant component (energy wasting).



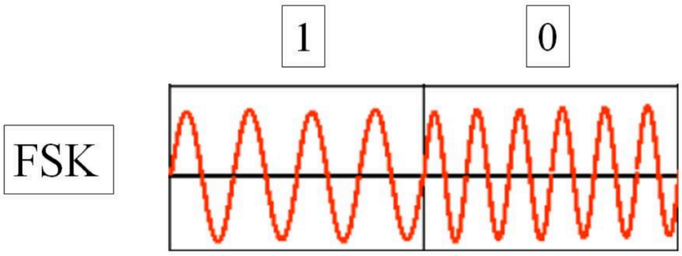
2. Frequency Shift Keying (FSK) modulation:

„0” bit: frequency increase by f_d (deviation) to the carrier frequency.
„1” bit: frequency decrease by f_d (deviation) to the carrier frequency.

$S_{(0)}(t) = A \cdot \sin((\omega_c + \omega_d) \cdot t)$

$S_{(1)}(t) = A \cdot \sin((\omega_c - \omega_d) \cdot t)$

In this case the bit rate and the baud is equal. The necessary spectral width is: $2 \cdot \omega_d$
It is easy to demodulate at the receiver. Disadvantage is the high frequency range.

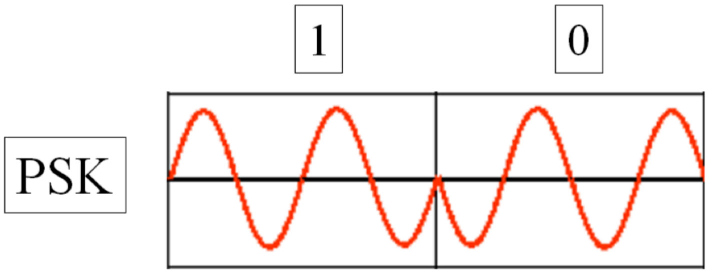


3. Phase Shift Keying (PSK) modulation:

„0” bit: inverse phase to the phase of the carrier signal.
„1” bit: identical phase with the phase of the carrier signal.

$S_{(0)}(t) = A \cdot \sin(\omega_c \cdot t + \pi) = -A \cdot \sin(\omega_c \cdot t)$

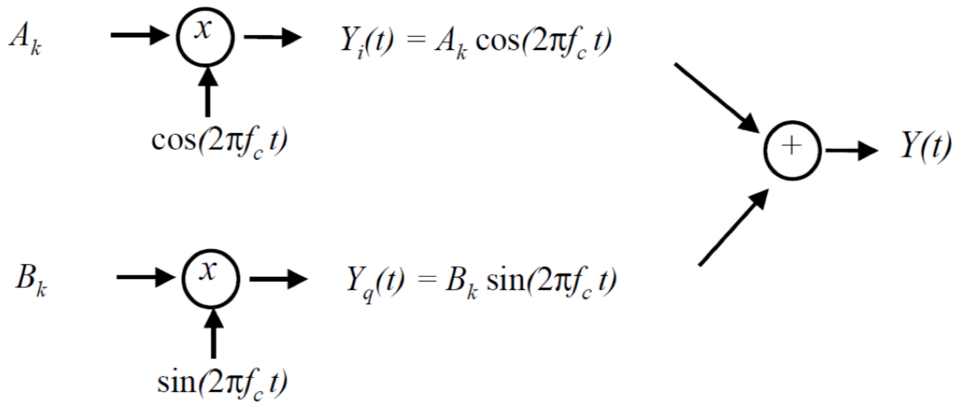
$S_{(1)}(t) = A \cdot \sin(\omega_c \cdot t)$



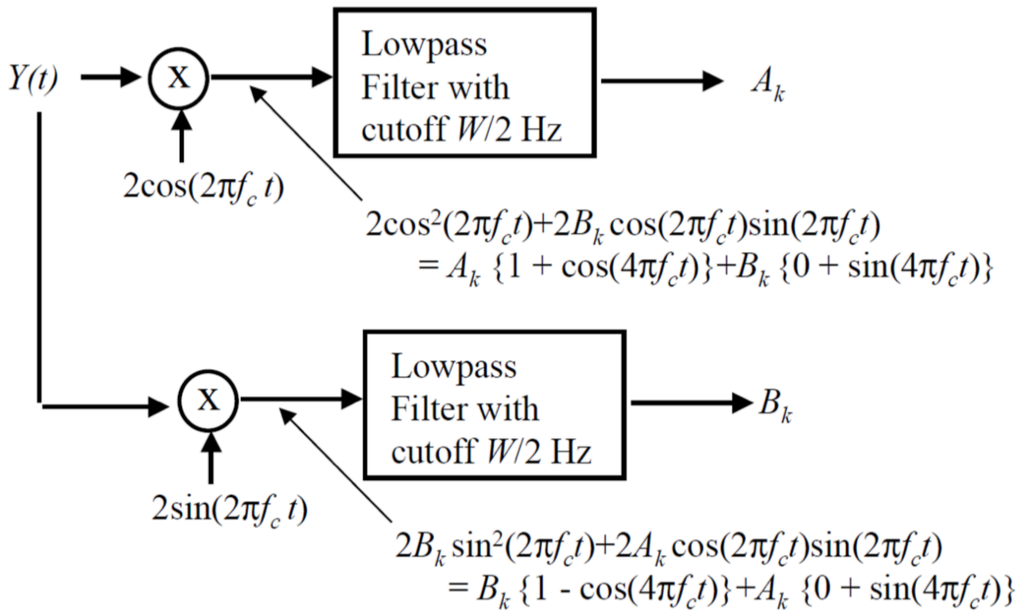
Multilevel PSK: In this case the bit rate and the baud is equal.
E.g.: 4 level PSK (Quadrate PSK, QPSK): phase displacements: 0º, 90º, 180º és 270º.
In this case two bits are transferred during one bit time.

4. Quadrature Amplitude Modulation (QAM): transmitter

Modulate $\cos(2\pi f_c t)$ and $\sin(2\pi f_c t)$ by multiplying them by A_k and B_k respectively for $(k\text{-}1)T < t < kT$:

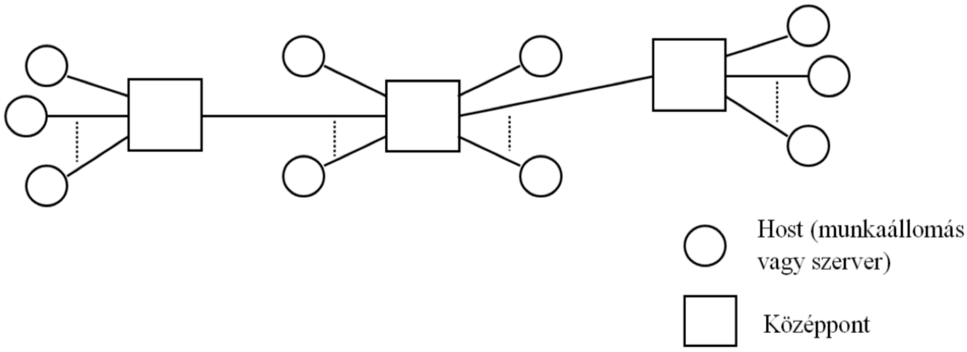


4. Quadrature Amplitude Modulation (QAM): receiver



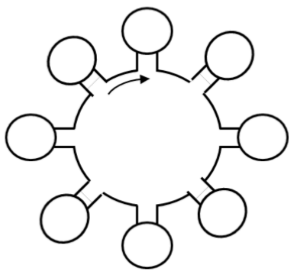
13) List four topology types and characterize them (text and figure).

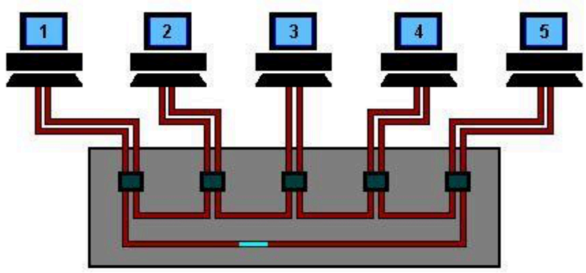
1. Star and extended star topology:



- Star: One central element (device) with end nodes.
- Extended star: Several central elements with end nodes. Extension is based on two levels, usually.
- Failure of a given central element affects the nodes and other central elements connected to this central element.

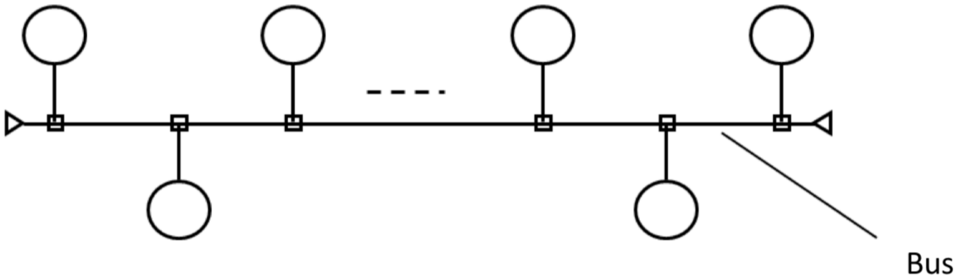
2. Ring topology:





- Each node has upside and downside neighbour.
- The ring has defined transmission direction.
- The concentrator increase the robustness, but it is a single point of failure element.
- The frame sent by the transmitted is received back from the upside neighbour and deleted from the ring by the source node.
- Secondary ring may be applied in special cases (see later).

3. Bus topology:



- Several nodes are connected to the same channel (wire).
- The bus may be unidirectional or bidirectional.
- Any physical element of the wire may be single point of failure element.
- Discontinuity of the media creates reflection of the signal in that physical point. Multiple reflections produce noise.

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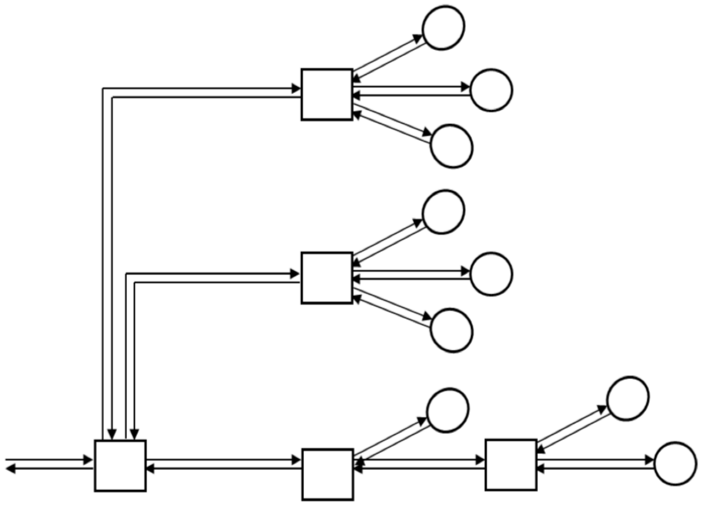
4. Tree topology:



- Is an extension of the extended star topology. The number of hierarchies is not limited. In practice the number of intermediate nodes and end nodes is finite.
- Any two leaves can communicate through a single path. No redundancy of the path exists.
- May appear different transmission intensities of the PDUs at different regions of the network.

14) List static and dynamic channel access mechanisms and characterize them.

1. Static channel access methods:

Frequency Division Multiple Access (FDMA):
- The channel is split into subchannels consisting disjunctive frequency ranges to reduce the channel access contention. In idea situation each source is mapped to different subchannels eliminating the contention.

Time Division Multiple Access (TDMA):
- The access to the common channel is scheduled in consecutive time intervals (slots) for the sources in the contention. Each source has own time slot. Just one source sends in a given time slot.

Wavelength Division Multiple Access (FDMA):
- Similar to the FDMA, but the media is fibre cable or EM space, and the signal is light.

2. Dynamic channel access methods:

- Sending without carrier sensing

- Time slotting

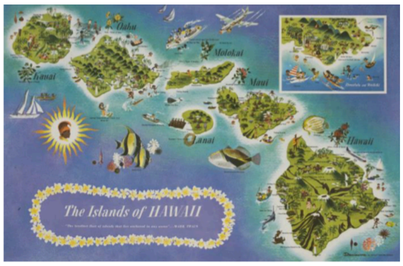
- Carrier Sense Multiple Access (CSMA)

- Collision Detection

- Token Based Access

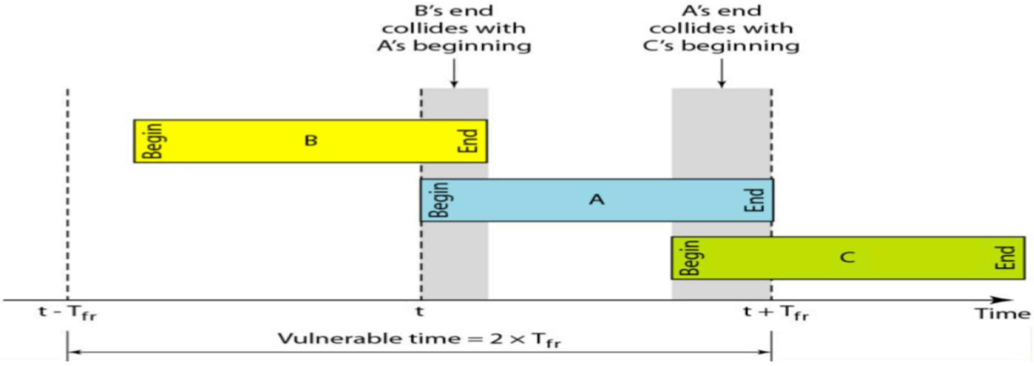
- Code Division Multiple Access

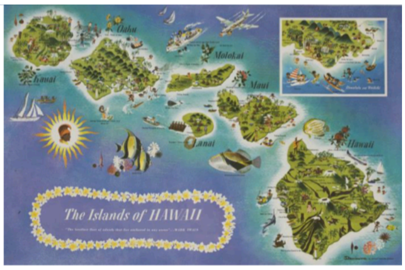
15) Similarities and differences of the ALOHA and Slotted ALOHA communication mechanisms (text and figure).



ALOHA:

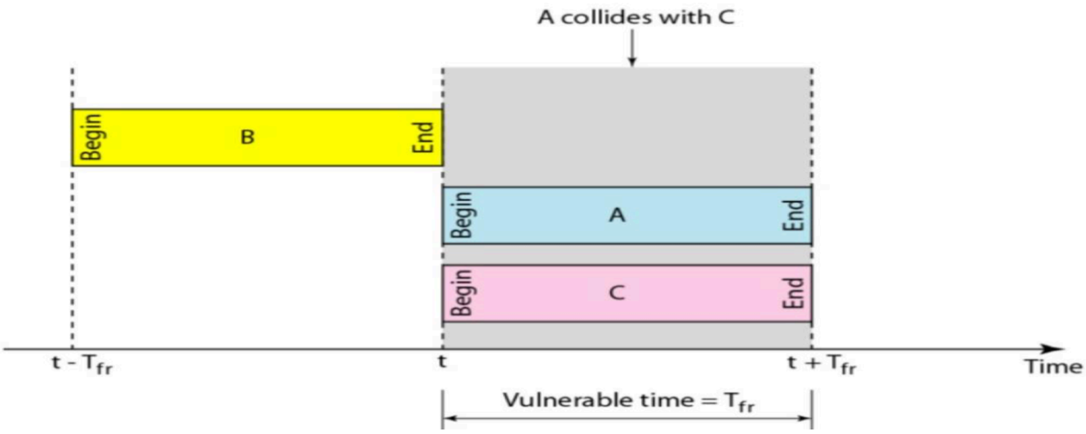
- Simplest channels access without carrier sensing.
- Source: University of Hawaii – radio channel communication between islands.
- Frame is sent on the common channel.
- Receiver send acknowledgement if no collision.
- Waiting for a random time interval and retransmission if collision is detected by the source.
- Simple to implement, simple to execute.
- Maximum efficiency of the channel usage < 18 %.





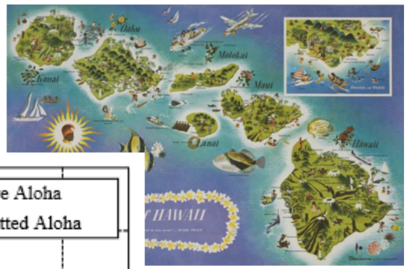
Slotted ALOHA:

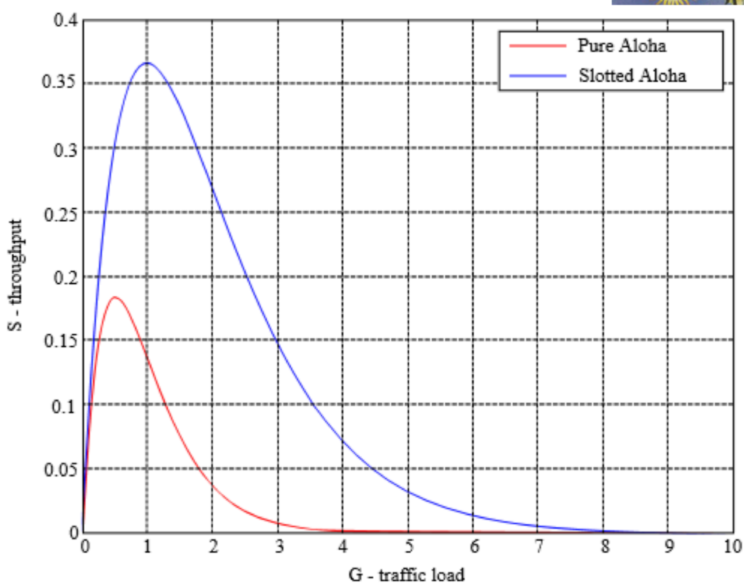
- Extension of ALOHA.
- Usage of time slot: frame transmission time
- Receiver send acknowledgement if no collision.
- Waiting for a random time interval and retransmission if collision is detected by the source.
- Maximum efficiency of the channel usage < 37 %.



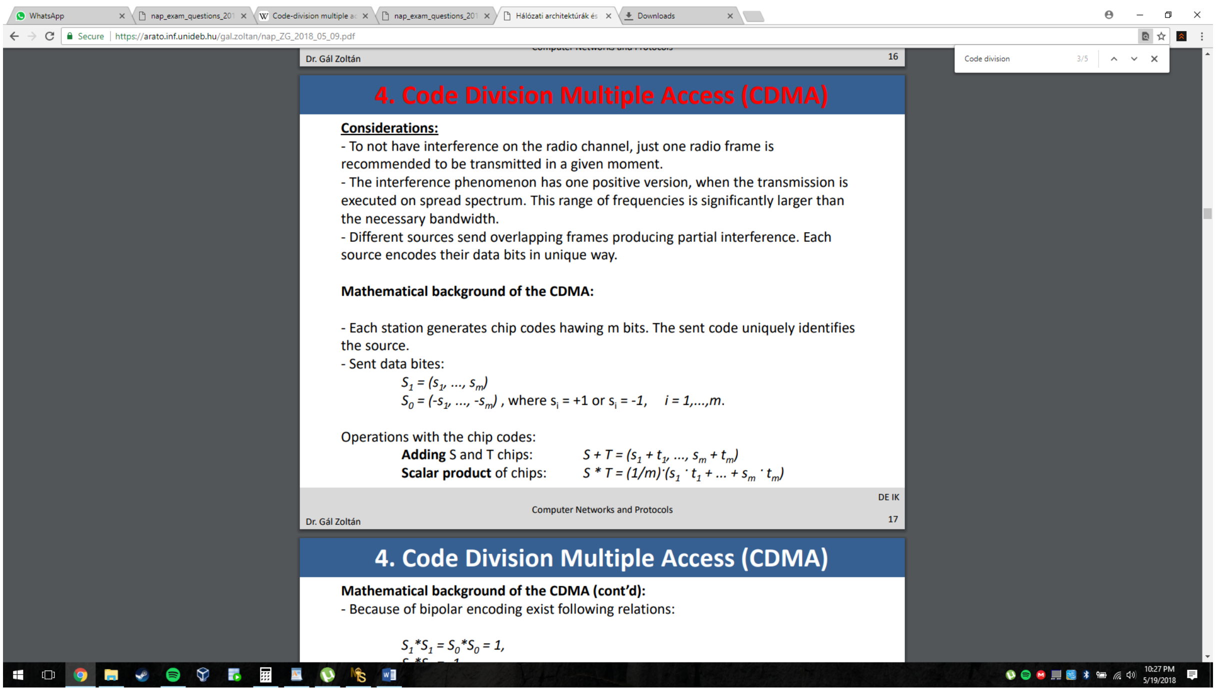
Channel usage efficiency of ALOHA and S-ALOHA:

- Distribution law of the sending trials is Poisson.





16) Describe the Code Division Multiple Access mechanism



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4. Code Division Multiple Access (CDMA)

Considerations:

- To not have interference on the radio channel, just one radio frame is recommended to be transmitted in a given moment.
- The interference phenomenon has one positive version, when the transmission is executed on spread spectrum. This range of frequencies is significantly larger than the necessary bandwidth.
- Different sources send overlapping frames producing partial interference. Each source encodes their data bits in unique way.

Mathematical background of the CDMA:

- Each station generates chip codes hawing m bits. The sent code uniquely identifies the source.
- Sent data bites:

$S_1 = (s_1, \ldots, s_m)$

$S_0 = (-s_1, \ldots, -s_m)$, where $s_i = +1$ or $s_i = -1$, $i = 1,\ldots,m$.

Operations with the chip codes:

Adding S and T chips: $S + T = (s_1 + t_1, \ldots, s_m + t_m)$

Scalar product of chips: $S * T = (1/m) \cdot (s_1 \cdot t_1 + \ldots + s_m \cdot t_m)$

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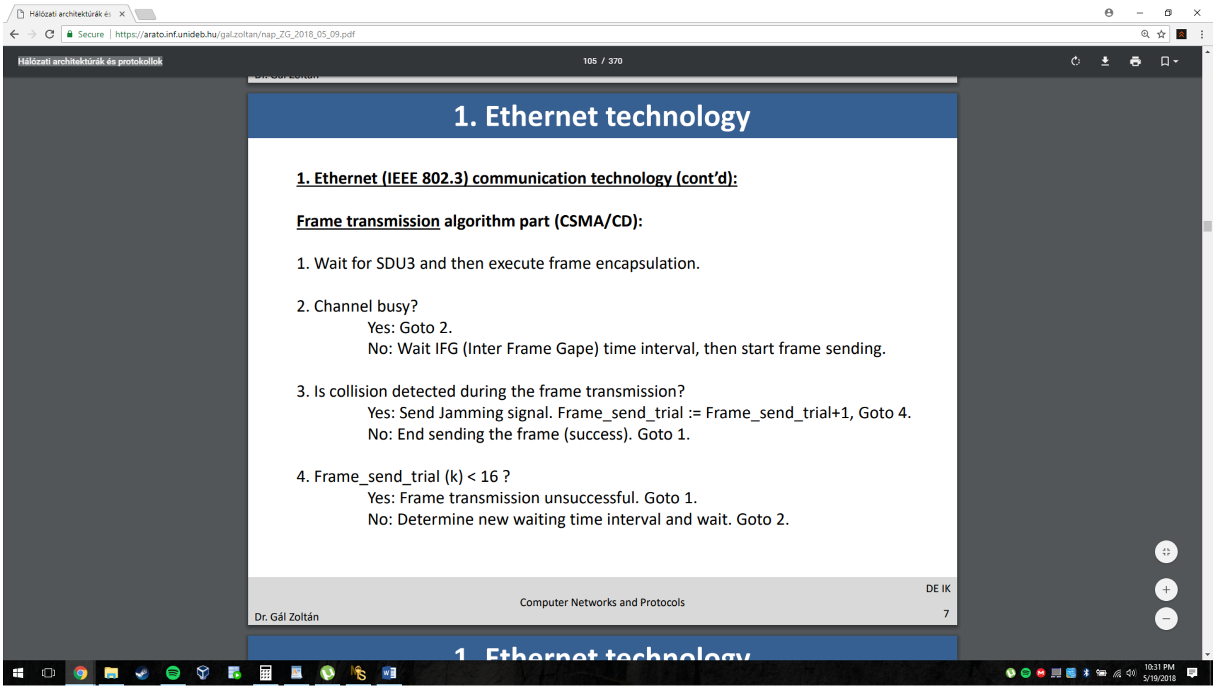
4. Code Division Multiple Access (CDMA)

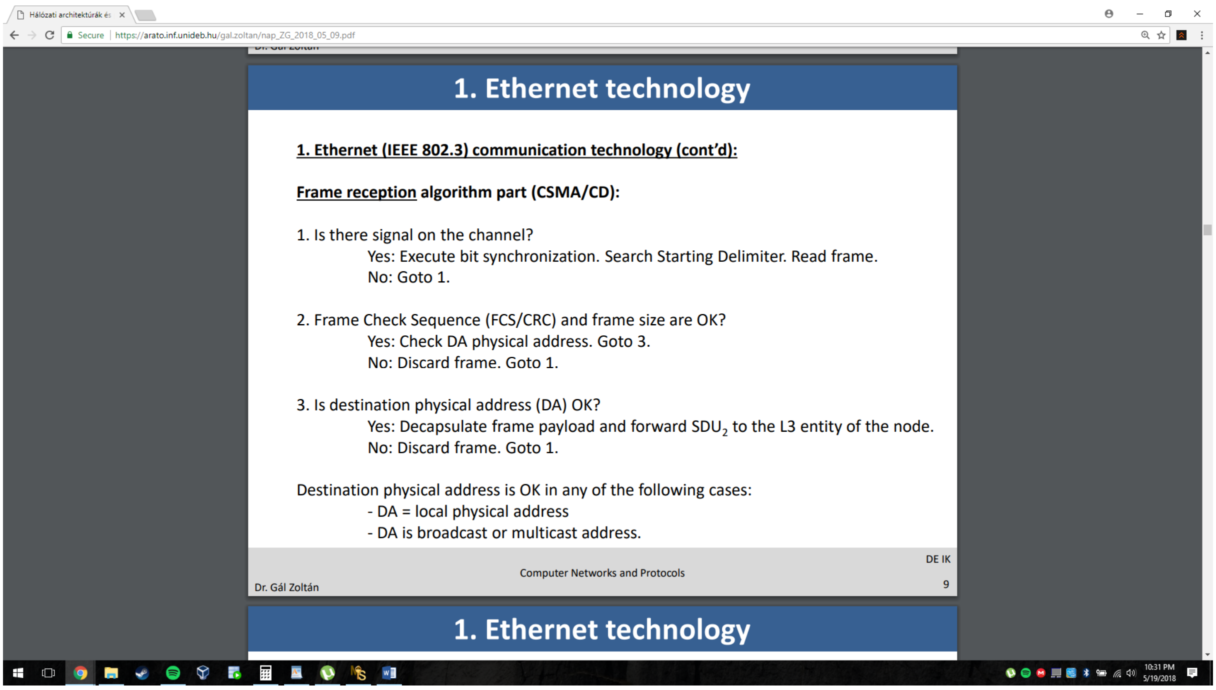
Mathematical background of the CDMA (cont'd):

- Because of bipolar encoding exist following relations:

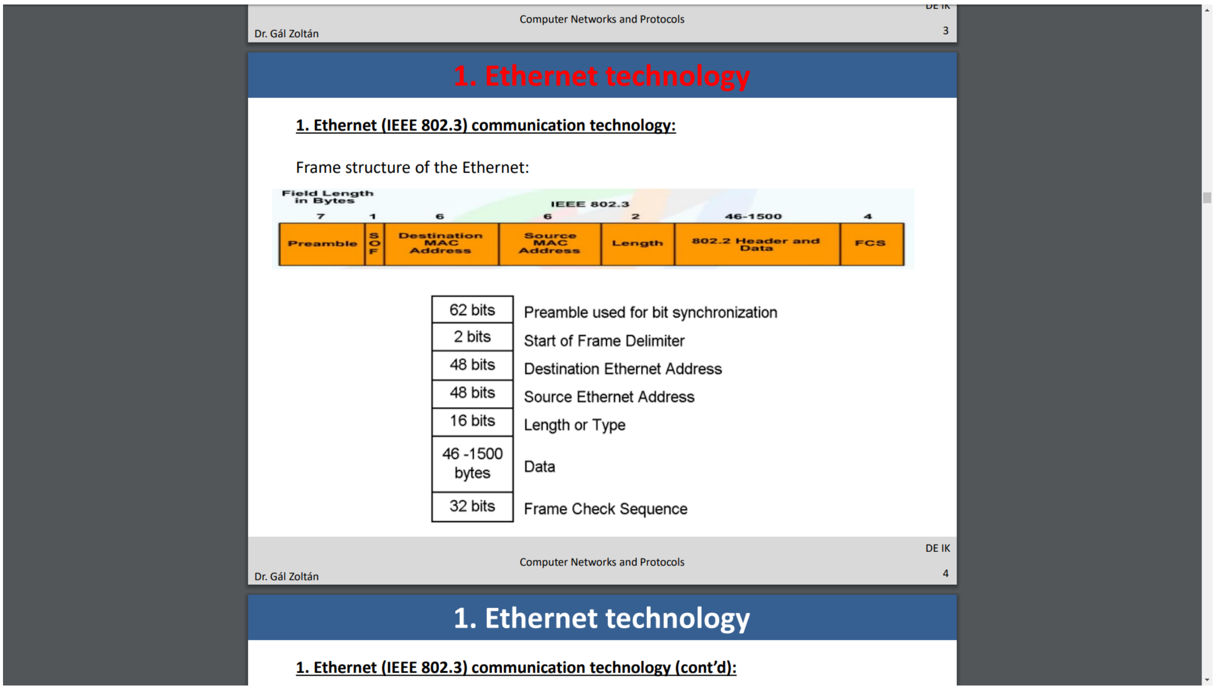
$S_1 * S_1 = S_0 * S_0 = 1,$

17) Description of the IEEE 802.3 MAC algorithm (text and figure).



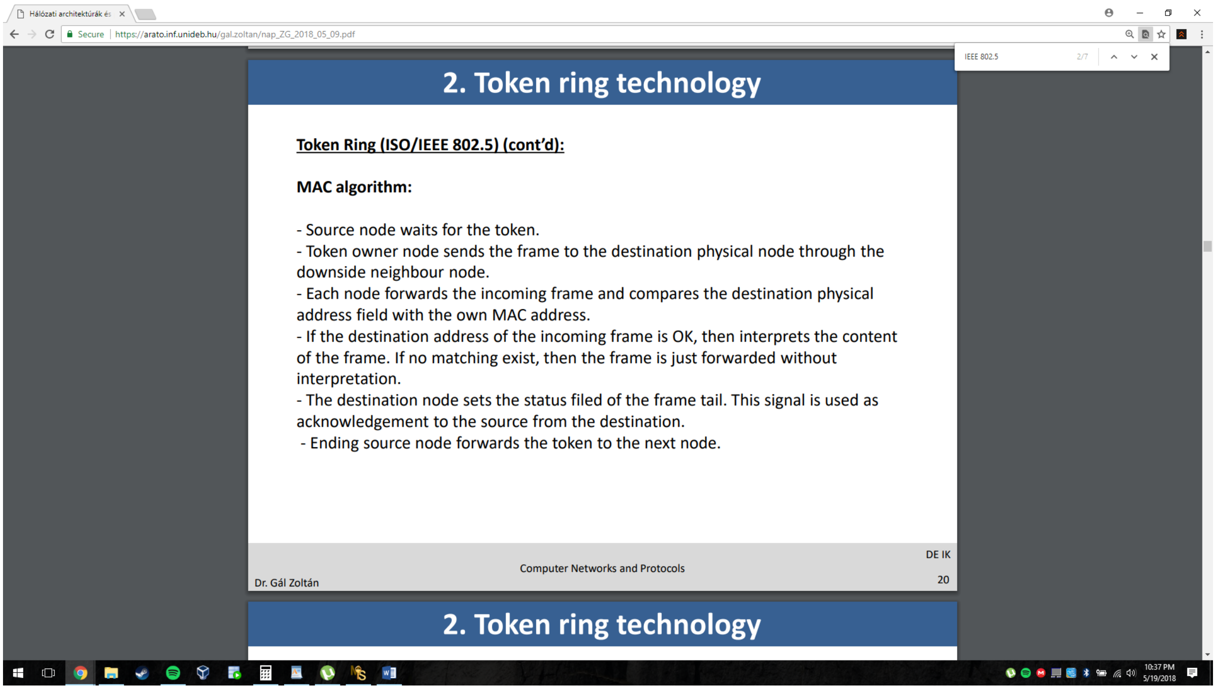


18) Description of the IEEE 802.3 MAC frame structure (text and figures).



19) Compare different Ethernet technology versions .

20) Description of the IEEE 802.5 MAC Algorithm (text and figure).



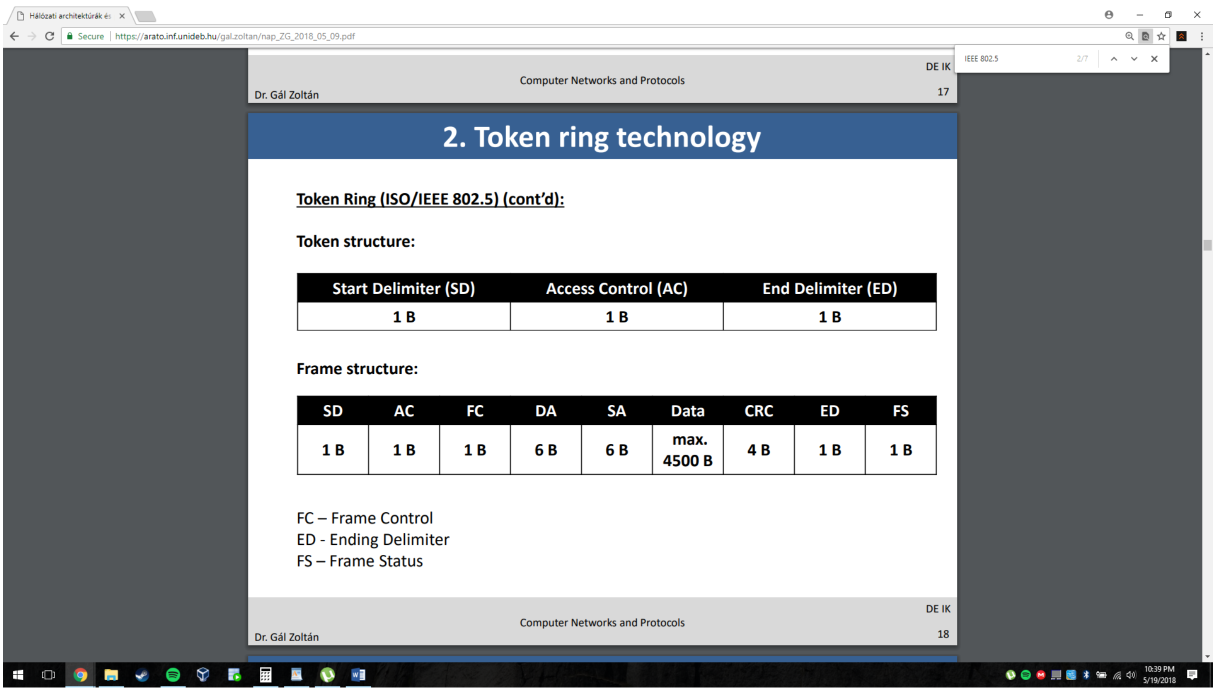
2. Token ring technology

Token Ring (ISO/IEEE 802.5) (cont'd):

MAC algorithm:

- Source node waits for the token.
- Token owner node sends the frame to the destination physical node through the downside neighbour node.
- Each node forwards the incoming frame and compares the destination physical address field with the own MAC address.
- If the destination address of the incoming frame is OK, then interprets the content of the frame. If no matching exist, then the frame is just forwarded without interpretation.
- The destination node sets the status filed of the frame tail. This signal is used as acknowledgement to the source from the destination.
- Ending source node forwards the token to the next node.

21)Description of the IEEE 802.5 MAC frame structure (text and figures).



2. Token ring technology

Token Ring (ISO/IEEE 802.5) (cont'd):

Token structure:

Start Delimiter (SD)	Access Control (AC)	End Delimiter (ED)
1 B	1 B	1 B

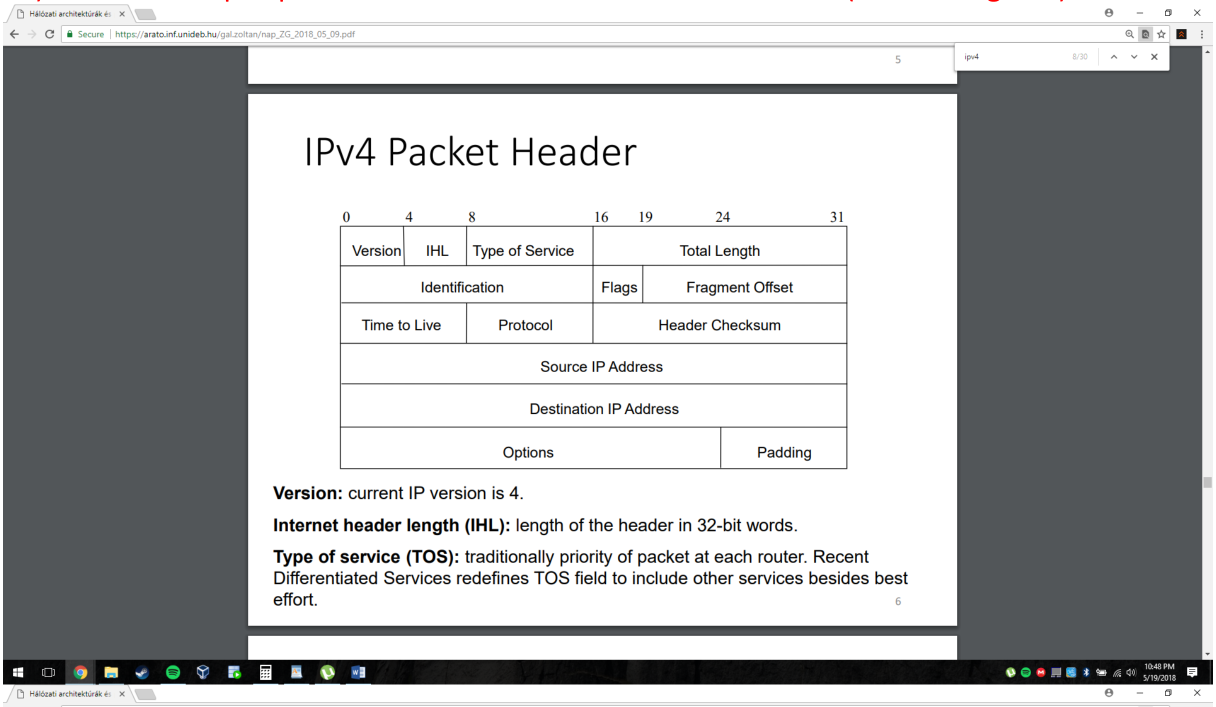
Frame structure:

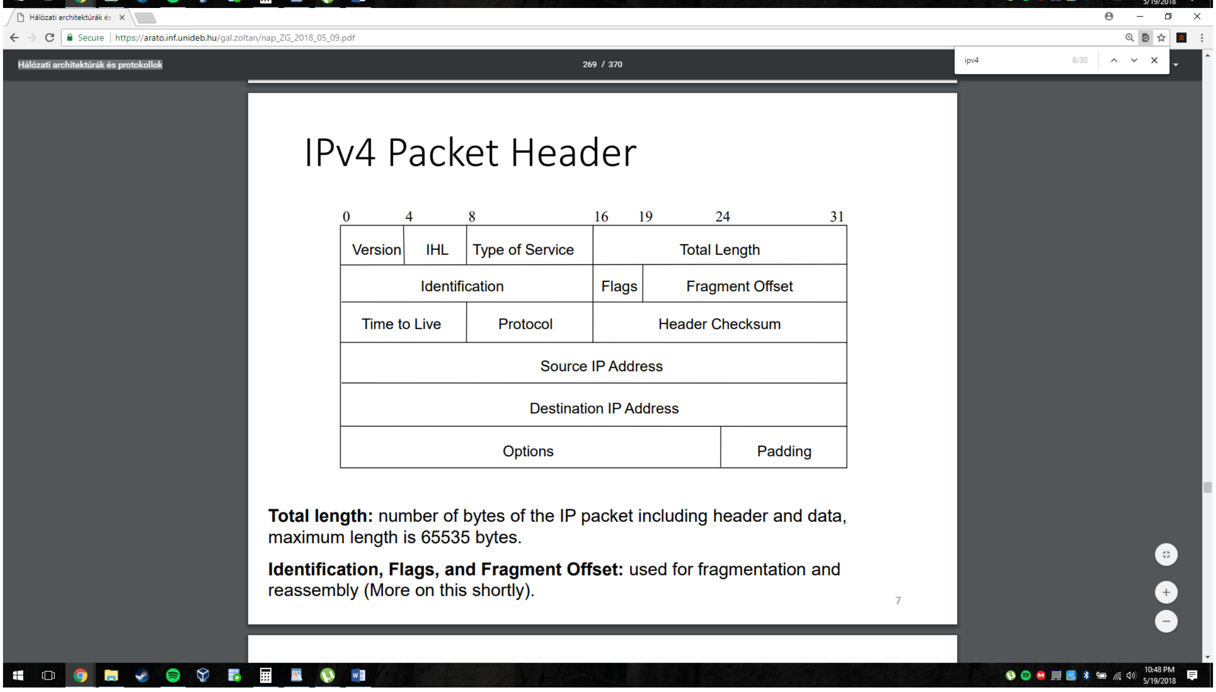
SD	AC	FC	DA	SA	Data	CRC	ED	FS
1 B	1 B	1 B	6 B	6 B	max. 4500 B	4 B	1 B	1 B

FC – Frame Control
ED - Ending Delimiter
FS – Frame Status

22) Compare the datagram characteristics on L2, L3 , L4 and give concrete communication technology example for each level (text and figures).

23)Describe the ipv4 packet structure and functions of the fields (text and figures).





IPv4 Packet Header

0	4	8	16	19	24	31
Version	IHL	Type of Service	Total Length			
Identification			Flags	Fragment Offset		
Time to Live		Protocol	Header Checksum			
Source IP Address						
Destination IP Address						
Options					Padding	

Time to live (TTL): number of hops packet is allowed to traverse in the network.

- Each router along the path to the destination decrements this value by one.
- If the value reaches zero before the packet reaches the destination, the router discards the packet and sends an error message back to the source.

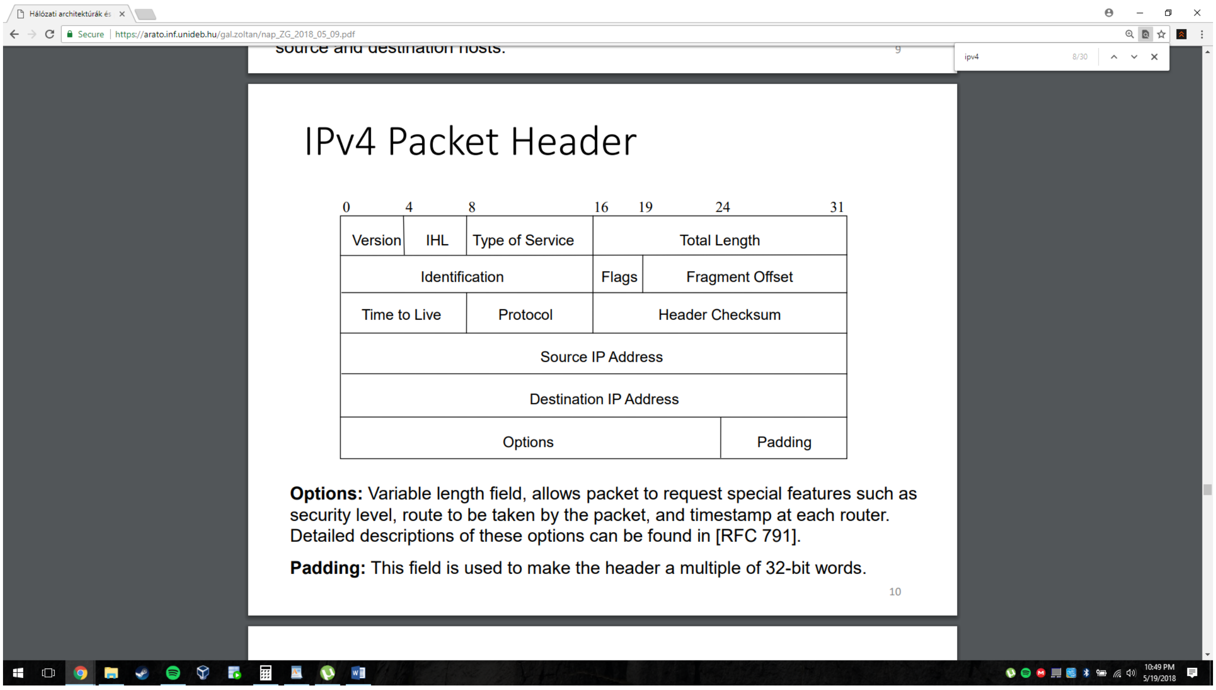
IPv4 Packet Header

0	4	8	16	19	24	31
Version	IHL	Type of Service	Total Length			
Identification			Flags	Fragment Offset		
Time to Live		Protocol	Header Checksum			
Source IP Address						
Destination IP Address						
Options					Padding	

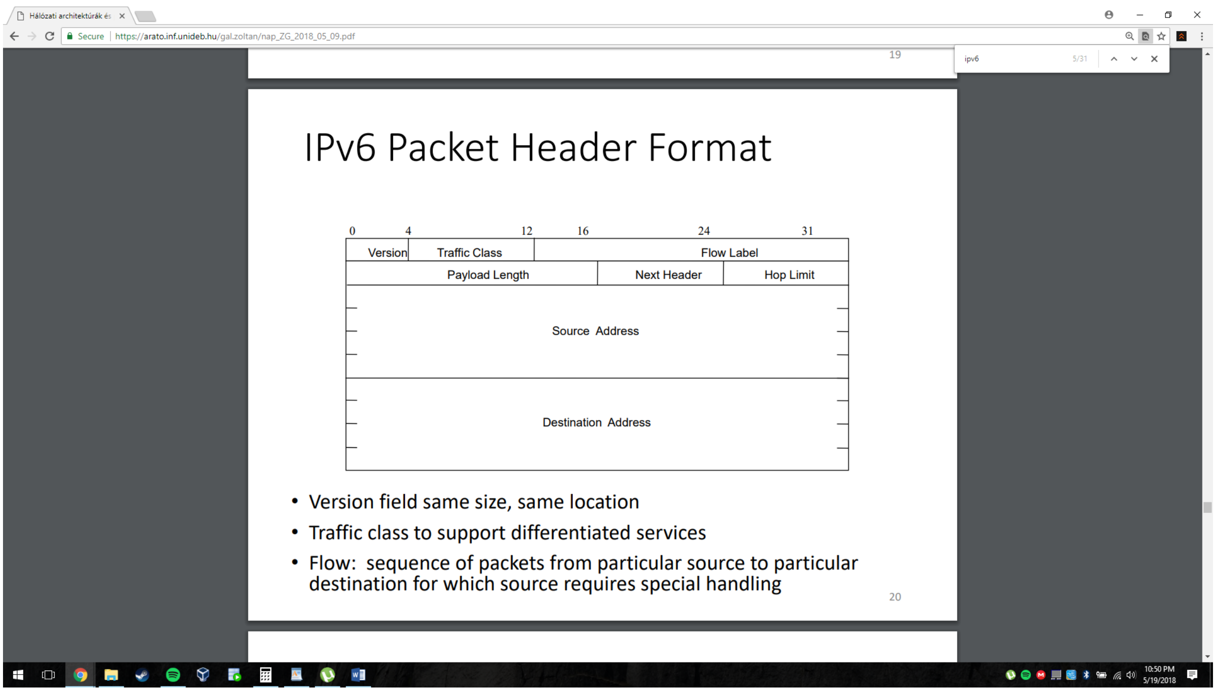
Protocol: specifies upper-layer protocol that is to receive IP data at the destination. Examples include TCP (protocol = 6), UDP (protocol = 17), and ICMP (protocol = 1).

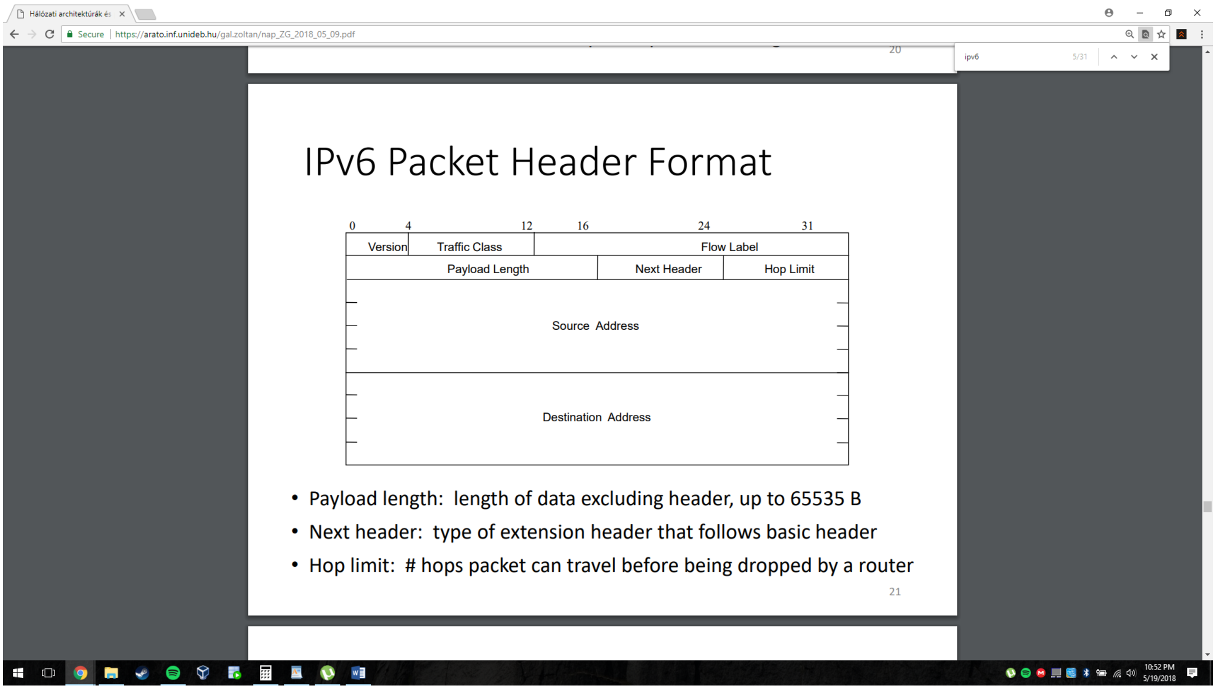
Header checksum: verifies the integrity of the IP header.

Source IP address and **destination IP address:** contain the addresses of the source and destination hosts.

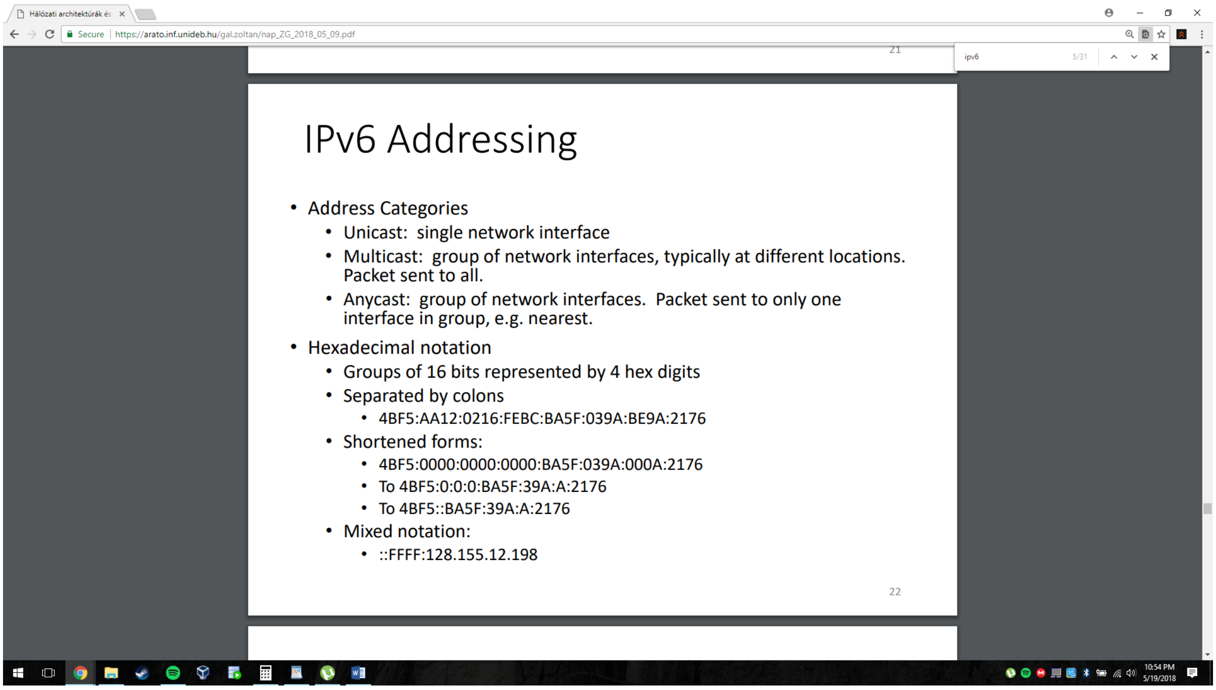


24) Describe ipv6 packet structure and functions of the fields (text and figures).

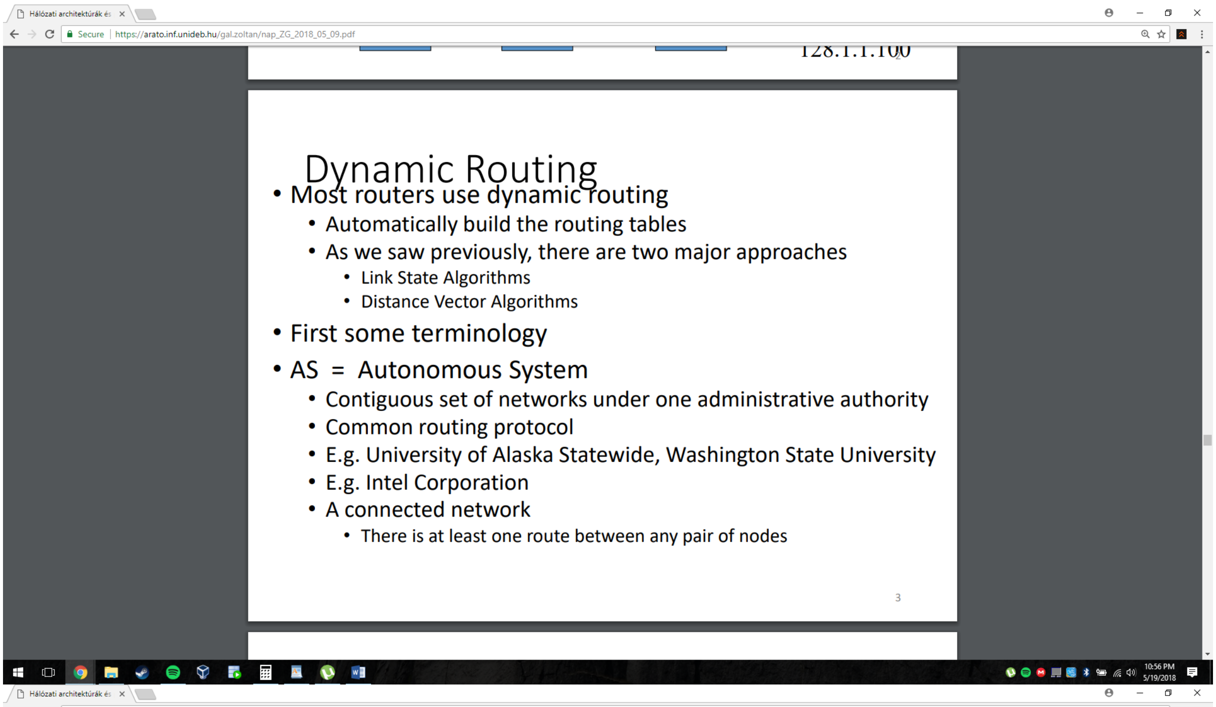


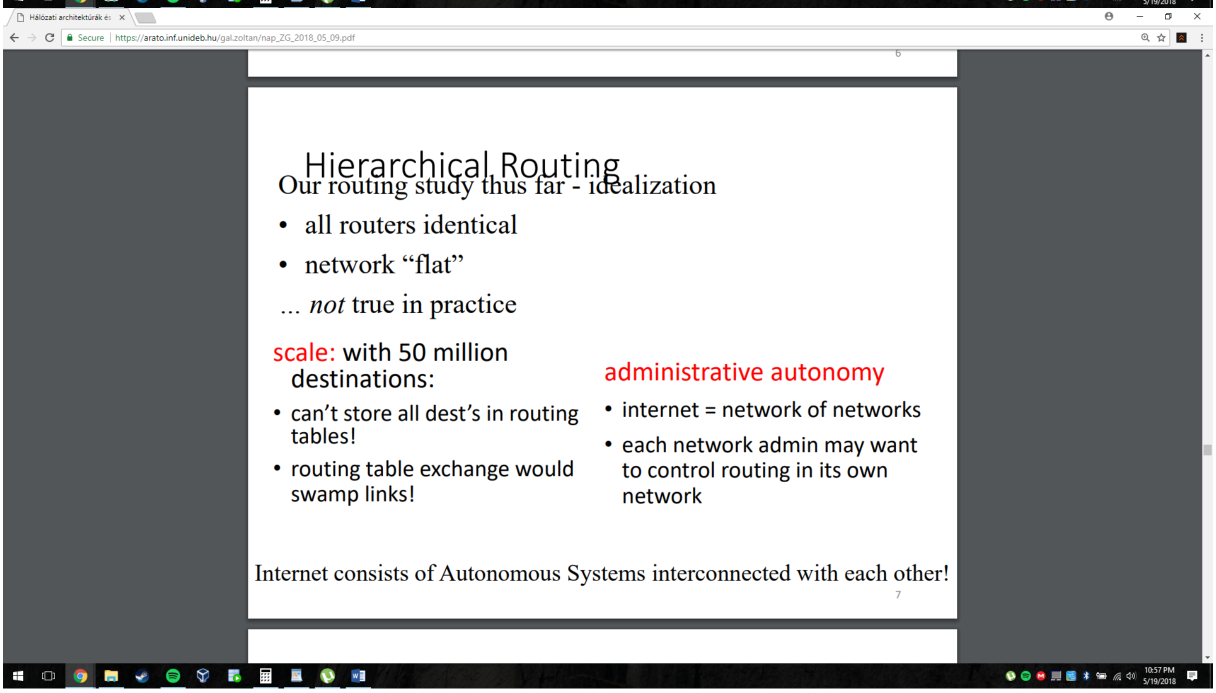


25)Characterize the IPv6 address types with examples (text).



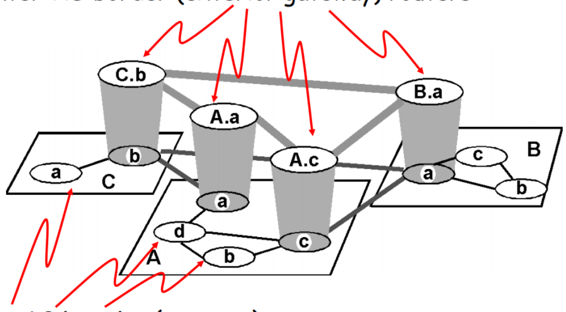
26) List routing protocol groups and give examples of them.





Internet AS Hierarchy

Inter-AS border (exterior gateway) routers



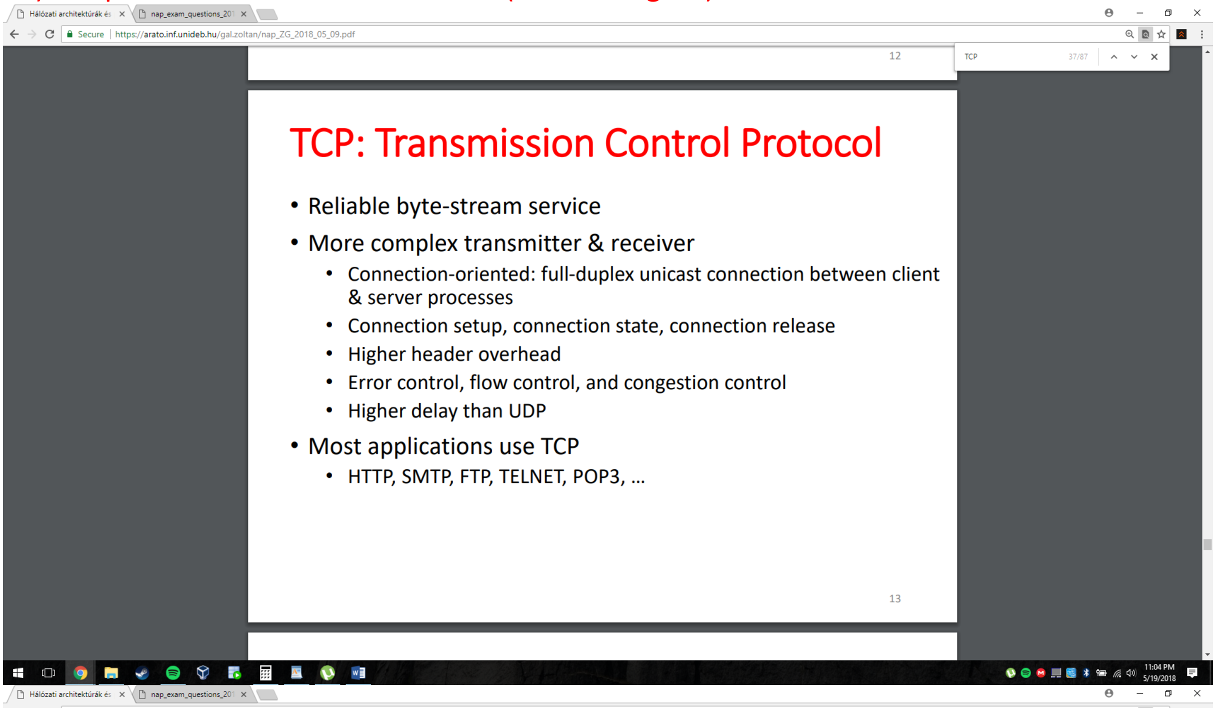
Intra-AS interior (gateway) routers

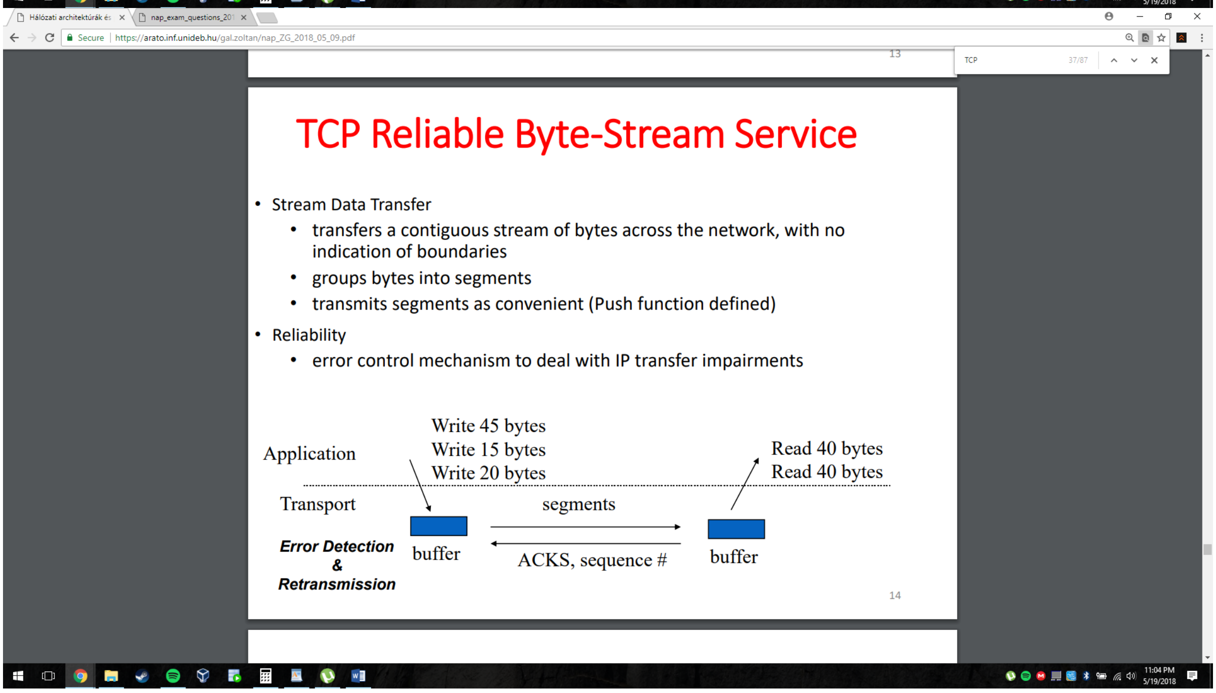
8

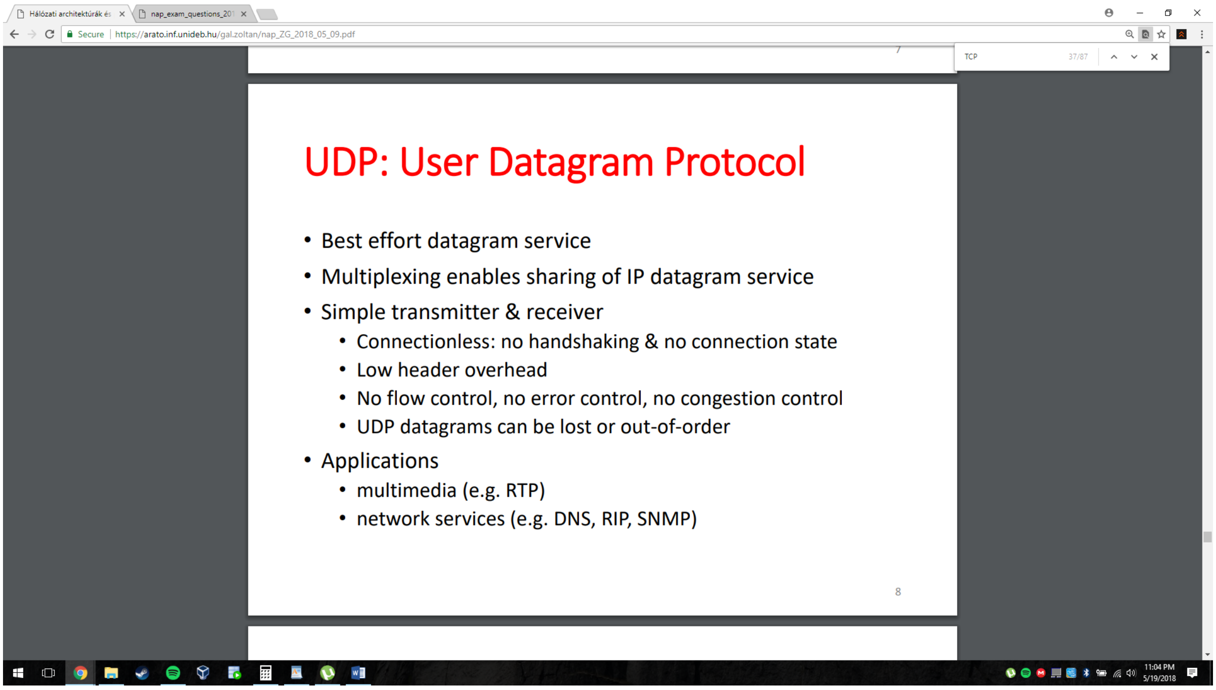
Intra-AS Routing

- Also known as Interior Router Protocols (IRP) or Interior Gateway Protocols (IGP)
- Most common:
  - RIP: Routing Information Protocol
  - OSPF: Open Shortest Path First
  - IGRP: Interior Gateway Routing Protocol (Cisco proprietary)

27)Compare TCP and UDP mechanism (text and figure).



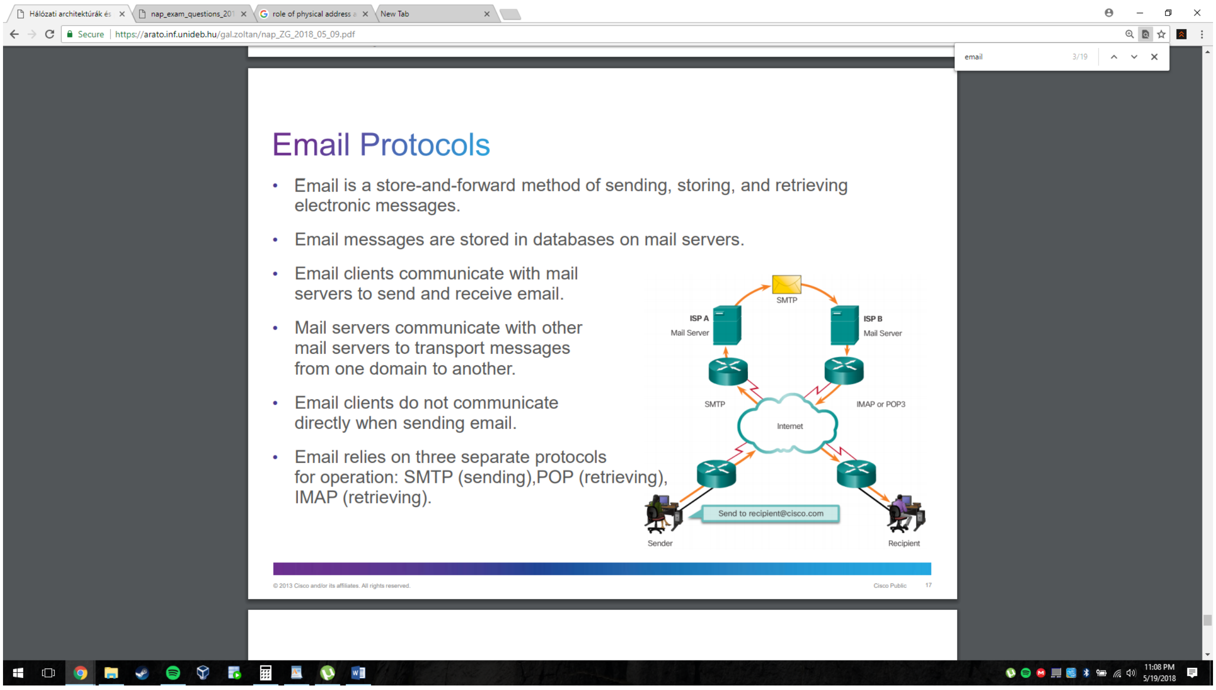




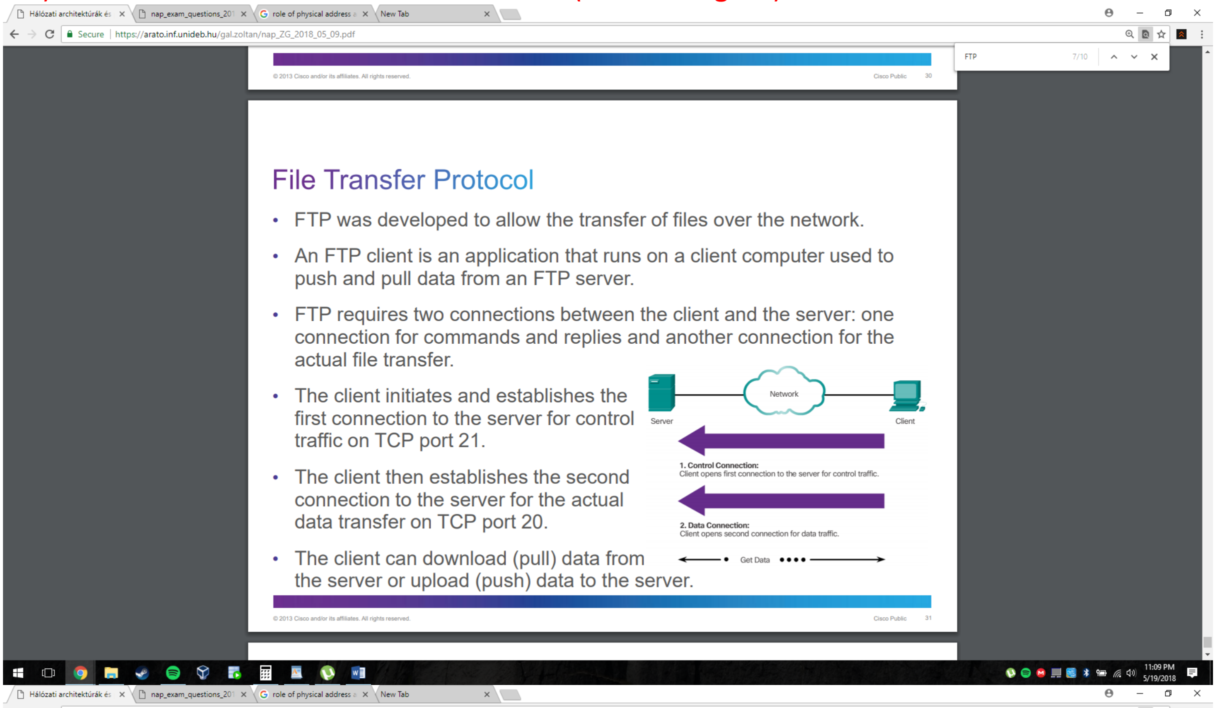
28) explain the role of the physical address ,logical address , port identifier

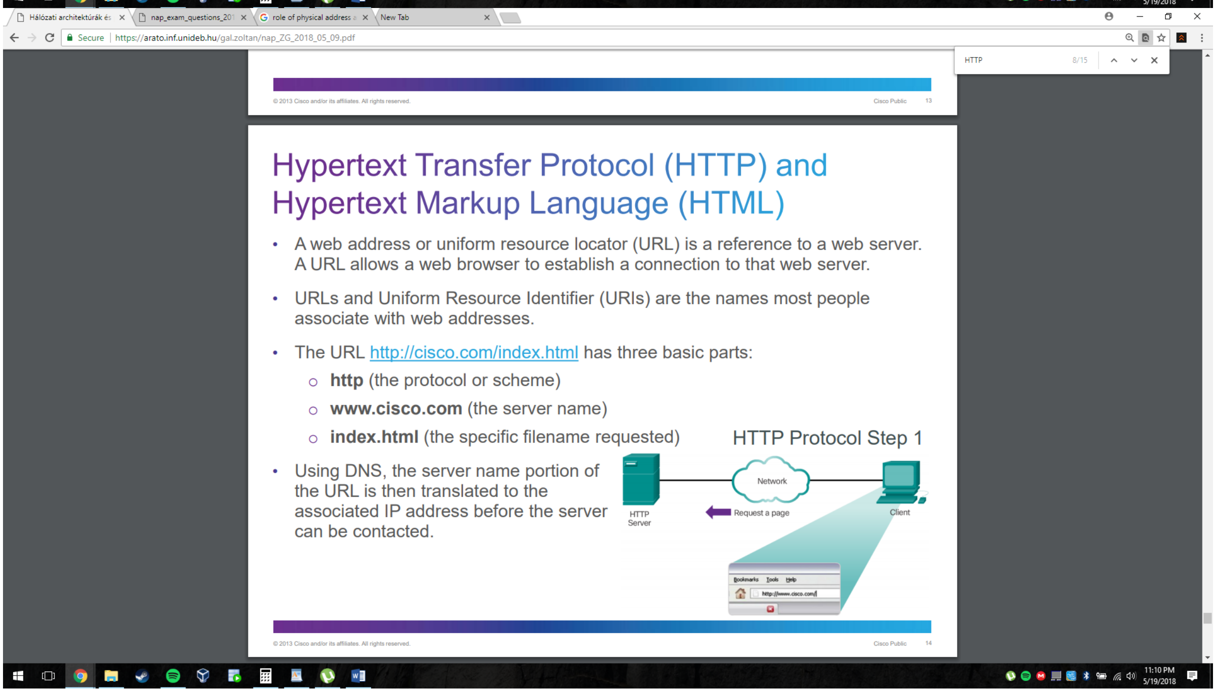
The **physical address** is the local **address** of a node; it is used by the data link layer to deliver data from one node to another within the same **network**. The **logical address** defines the sender and receiver at the **network** layer and is used to deliver messages across multiple **networks**.

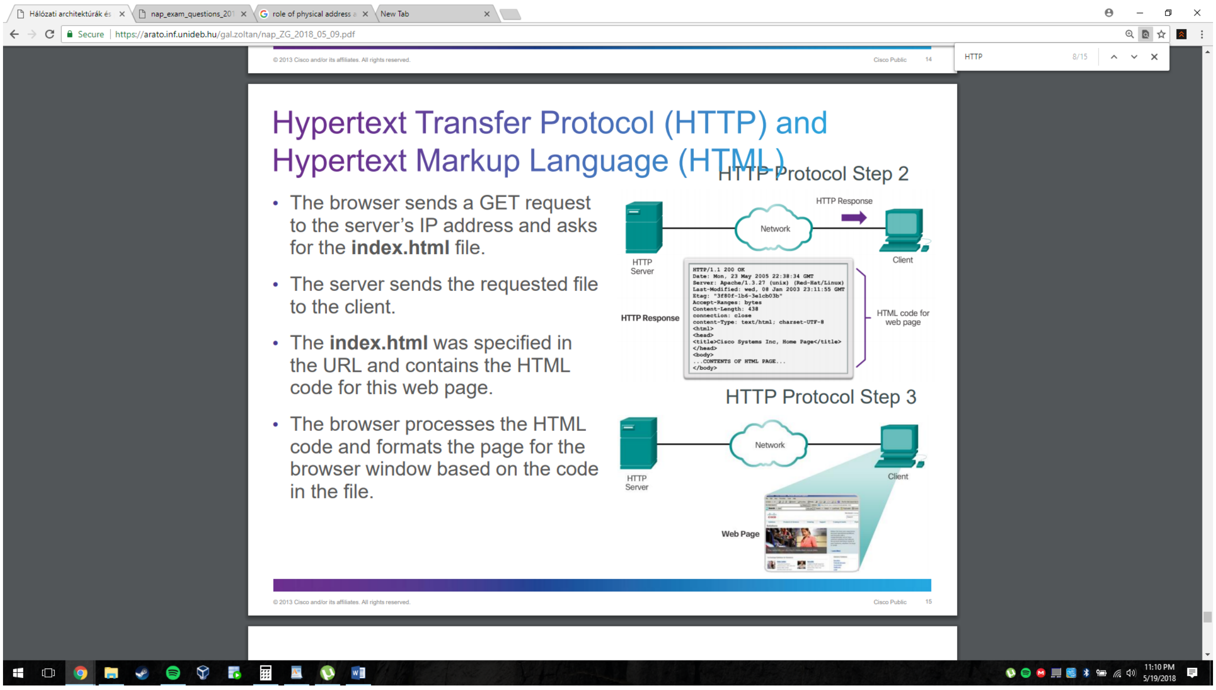
29) Describe the email service (text and figure).



30) Describe the FTP and HTTP mechanism (text and figure).






Hypertext Transfer Protocol (HTTP) and Hypertext Markup Language (HTML)
The browser sends a GET request to the server's IP address and asks for the index.html file.
The server sends the requested file to the client.
The index.html was specified in the URL and contains the HTML code for this web page.
The browser processes the HTML code and formats the page for the browser window based on the code in the file.
HTTP Protocol Step 2
HTTP Response
Network
HTTP Server
Client
HTML code for web page
HTTP Protocol Step 3
Web Page
https://arato.inf.unideb.hu/gal.zoltan/nap_ZG_2018_05_09.pdf
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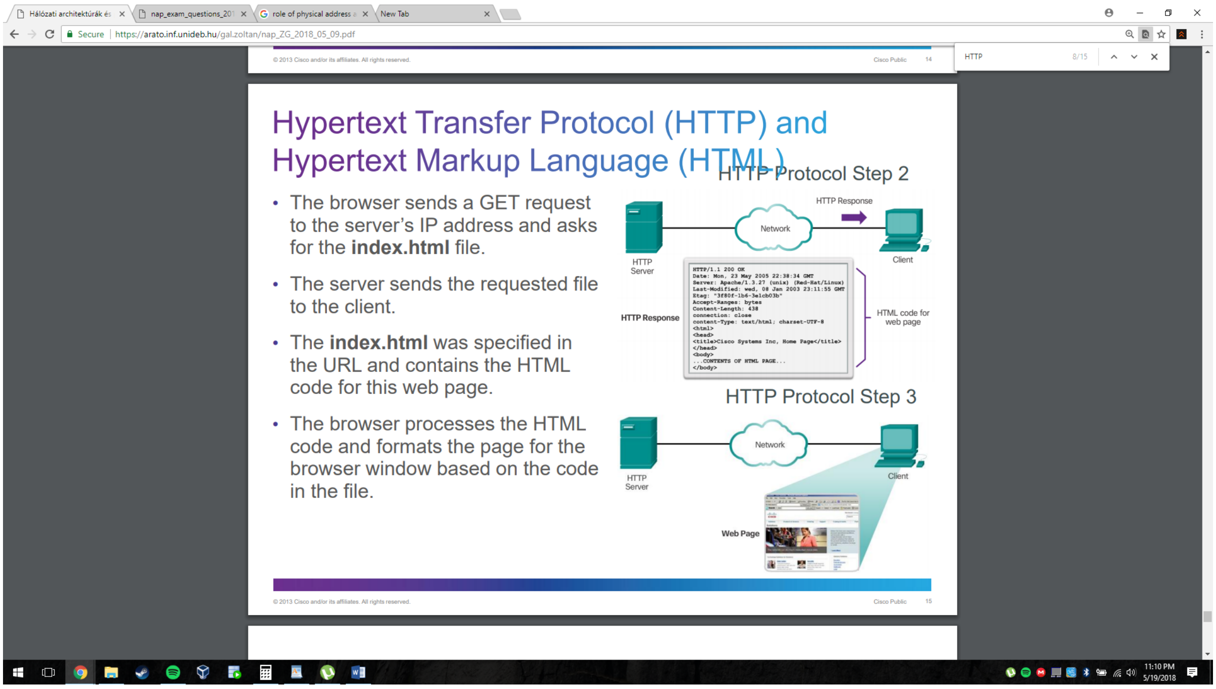
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HTTP Protocol Step 2
HTTP Response
Network
HTTP Server
Client
HTML code for web page
HTTP Protocol Step 3
Web Page
https://arato.inf.unideb.hu/gal.zoltan/nap_ZG_2018_05_09.pdf
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